Leakage makes every problem look equally easy: A multi-dataset measurement of the resampling-before-splitting artefact

Author names withheld for review
Affiliations withheld for review

Abstract

Applying synthetic over-sampling before rather than after the train-test split is among the most frequently reported causes of irreproducible results in applied machine learning. Recent large-scale work has begun to quantify leakage across many mechanisms and datasets; we focus narrowly on the distinctive statistical signature of resampling-before-splitting and on whether it can be recognised from reported results. We measure the end-to-end artefact on 41 public binary classification datasets spanning imbalance ratios from 1.02 to 109.9, with seven classifier families, four over-sampling methods and ten random seeds. The mean inflation of minority-class F1 is 0.224 (descriptive 95% bootstrap interval 0.220 to 0.228). More consequentially, leakage does not simply raise scores: it destroys the relationship between score and problem difficulty. Under the correct ordering, F1 varies with imbalance, dimensionality and class overlap as expected; under the leaked ordering it collapses onto a narrow band set mainly by the classifier's capacity to memorise interpolated near-duplicates, with 4.86 times less variance. The premium is therefore almost entirely predicted by how hard the problem actually is ($R^2 = 0.795$). The leaked score is not merely insensitive to difficulty; it moves against it, rising with the imbalance ratio (r = 0.578) exactly where the honest score falls (r = -0.268), because a more imbalanced problem requires more synthetic points and also changes the composition of the resampled test partition. A controlled simulation separates factors that co-vary in real data, while a contamination measure traces one mechanism contributing to the artefact. Two practical results follow. Over-sampling to a minority prevalence of 0.25 rather than to full balance cuts the artefact by 25% and lowers the contamination that produces it, so partial resampling is a cheap partial safeguard. And because the artefact has a predictable signature, a reported result can be screened without access to the authors' code: a diagnostic trained on our corpus and validated leave-one-dataset-out separates leaked from honest results with AUC 0.795 using only the reported F1, imbalance ratio, minority count, feature count and model family. We release it as an open-source tool.

Keywords: data leakage; class imbalance; SMOTE; synthetic over-sampling; empirical evaluation; reproducibility; benchmark study

1 Introduction

A machine learning result is a claim about a procedure applied to data that the procedure has not seen. Data leakage is the failure of that condition, and it is now recognised as a leading cause of results that do not replicate [1, 2, 3, 29, 30, 32]. Among its many forms, one is unusually easy

to commit and unusually easy to overlook: applying a synthetic over-sampler such as SMOTE [4] to a dataset and only then splitting it into training and test partitions. Two lines of code in the wrong order produce a pipeline that looks correct, runs without warning, and reports a number that describes something other than what the author believes it describes. The mechanism is not subtle once stated. SMOTE creates a new minority point by interpolating between a real minority point and one of its neighbours. If the interpolation happens before the split, some of those synthetic points inherit most of their coordinates from a real point that the split then assigns to the test set, while the synthetic point itself lands in the training set. The model does not have to generalise to those test cases. It has to remember them. That the error inflates results has been stated many times and demonstrated at length on a medical corpus [5]. A recent quantitative landscape has also measured oversampling leakage within a much broader taxonomy of leakage mechanisms [56]. What remains unresolved is whether this particular workflow error replaces the expected relationship between score and problem difficulty, how its end-to-end premium relates to contamination and test-distribution change, and whether its signature is stable enough to be recognised in results one did not produce. Those questions matter for practical reasons. A reviewer confronted with a reported F1 of 0.95 on a problem with two hundred minority cases has no way to act on a qualitative warning. An editor deciding whether to request code has nothing to weigh. A field trying to understand why its published numbers cluster in a suspiciously narrow range needs a quantitative account of why that would happen. This paper supplies the measurement. We assembled 41 public binary classification problems, 35 drawn from a standard benchmark collection [6, 7] and 6 from the employee-attrition literature, with imbalance ratios from 1.02 to 109.9, sample sizes from 208 to 5000 and between 3 and 180 predictors. On each we fitted the same model three ways, changing nothing but where the resampler sits: before the split, after the split on the training partition only, and not at all. Seven classifier families, four over-samplers and ten seeds give 11,200 matched leaked-versus-honest comparisons from 25,340 fitted models. The headline number is the smaller half of the finding. The larger half is a regularity we did not expect and have not found reported: leakage does not shift the score distribution upward so much as it flattens it. Under the correct ordering, F1 behaves as a measurement should, falling as the problem gets harder. Under the leaked ordering it stops responding to difficulty almost entirely, landing in a narrow band whose position depends mainly on which classifier was used. An easy problem and a hard problem, run through the same faulty pipeline, come out looking alike. This has a consequence worth stating plainly, because it inverts the usual intuition. The common description of leakage is that it makes results 'a bit too good'. That description is wrong in a way that matters: leakage does not add a bias to a measurement, it replaces the measurement with a different one. The reported quantity stops being a property of the problem and becomes a property of the model class. Three contributions follow.

- **C1** – Focused measurement. We characterise resampling-before-splitting as an end-to-end workflow artefact across a corpus spanning two orders of magnitude in imbalance, seven algorithm families and four over-sampling methods, emphasizing its variance collapse and reversal of the score–difficulty relationship rather than claiming the first measurement of leakage magnitude.
- **C2** – A regularity and its mechanism. We show that the leaked score is close to invariant to problem difficulty and, where it does respond, responds in the wrong direction; that the premium is therefore predictable from difficulty alone; and that both follow from a measurable contamination rate. A controlled simulation separates imbalance, dimensionality, class overlap and sample size, which real data confounds.
- **C3** – A screening tool. Because the artefact has a stable signature, a published result can be screened from what papers already report. We train such a screen, validate it leave-one-dataset-out, and release it as an installable package.

A companion study by the present authors measured this effect on a single attrition benchmark and concluded that replication on independent data was needed before the finding could be generalised [8]. The present work is that replication, at a scale that turns a case report into a measurement, and it asks the further questions the single-dataset design could not: whether the effect generalises, what governs its size, and whether it can be recognised from the outside.

2 Related work

2.1 Leakage

Kaufman and colleagues [1] gave the first systematic treatment, defining leakage as the introduction of information about the target that would not be legitimately available at prediction time, and cataloguing the forms it takes. Two decades later, Kapoor and Narayanan [2] collated twenty-two review papers covering seventeen scientific fields and identified leakage affecting at least 294 published studies, arguing that it is a principal driver of the reproducibility problem in machine-learning-based science. Whalen and colleagues [3] make the same case for genomics. Recent work has extended the evidence to workflow enforcement, public leaderboards, decentralised credit scoring and multivariate time-series classification [29, 30, 31, 32]. Most directly, Roth [56] reports 28 within-subject leakage experiments across 2,047 tabular datasets and includes oversampling-before-splitting among the memorisation mechanisms studied. That work establishes a broad quantitative landscape, including capacity and sample-size effects. Our narrower contribution is to characterise the resulting F1 distribution: its loss of sensitivity to difficulty, its reversal with imbalance, its relationship to a measurable contamination rate, and its detectability from quantities available to a reader. Vandewiele and colleagues [5] provide the closest detailed domain case, reproducing inflated preterm-birth results and showing that they collapse under correct evaluation.

2.2 Synthetic over-sampling and its sceptics

SMOTE [4] and its descendants Borderline-SMOTE [9] and ADASYN [10] are the most widely used response to class imbalance. Reviews document both the breadth of the resulting family and the persistent problems caused by overlap, noise and inappropriate synthetic samples [35, 36, 39, 44, 50]. A steady line of work questions whether rebalancing helps. Blagus and Lusa [11] showed that SMOTE is largely ineffective in high dimensions and can be harmful for some classifiers. Van den Goorbergh and colleagues [12] demonstrated that imbalance corrections destroy the calibration of risk prediction models while leaving discrimination unchanged, a result Carriero and colleagues [13] extended to machine learning models by simulation. Large empirical and theoretical comparisons likewise find that no rebalancing can be competitive when the classifier is appropriately selected or tuned [14, 40]. Our results are consistent with this line and add a specific observation: in our corpus the best-performing condition is frequently not to resample. Across the whole corpus, no resampling beat correctly-ordered resampling in 34.7% of matched comparisons.

2.3 Large-scale empirical benchmarking

The practice of evaluating a claim across many datasets rather than a favoured few goes back to Demsar [15]. PMLB [6, 7] curates a large collection of standardised classification and regression problems for exactly this purpose. Our design borrows the logic: the question is not whether the artefact appears in a particular case but how its magnitude behaves as a function of dataset properties, and that question needs a corpus wide enough to vary those properties.

2.4 The gap

Three things are missing from the literature above. First, no measurement of the effect size of resampling-before-splitting as a function of dataset characteristics. Second, no account of what the leaked score is actually measuring when it is not measuring the problem. Third, no way for a reader to act on the concern in a specific case. This paper addresses all three.

3 Method

3.1 Corpus

We assembled 41 binary classification problems in two groups. Group A consists of 35 benchmark problems derived from PMLB [6, 7]. PMLB distributes datasets in a harmonised numeric format with a documented provenance; where a source is multi-class we binarise it one-versus-rest on the rarest class with at least thirty cases, which is the standard construction used to build imbalanced benchmark collections from multi-class sources. Group B consists of 6 employee-attrition and turnover datasets. We chose attrition as a deliberately coherent applied case study rather than as an additional convenience sample: it is a canonical imbalanced decision problem, resampling is common in its recent predictive literature, multiple studies reuse the same public benchmarks, and false-positive as well as false-negative errors carry direct resource implications. The HR datasets also anchor the mild-imbalance end of the corpus (ratios 1.02 to 7.42), allowing that applied domain to be located within rather than substituted for the broader benchmark landscape. One dataset [16] is survey data collected for the purpose rather than a re-publication of an existing corpus. Two filters apply. A dataset enters only if it has at least thirty minority cases, because below that a 70:30 split leaves too few test positives for F1 to be stable. Datasets larger than 5,000 rows are stratified-subsampled to 5,000. The cap standardises the maximum effective sample size across the factorial, prevents a small number of large problems from dominating computational cost, and makes repeated SMOTE neighbour searches, kernel-SVM fits and ten-seed comparisons feasible under a common budget while retaining each dataset's class ratio. It also defines the scope of the estimand: the results describe capped tabular problems, not enterprise-scale populations. Because leakage magnitude can vary with sample size [56], extrapolation beyond 5,000 observations requires a dedicated within-dataset scaling analysis. Both thresholds are exposed as parameters in the released code.

Table 1: The corpus. IR is the ratio of majority to minority cases after any subsampling. 'Construction' records how a binary target was obtained: benchmark problems that were already binary are marked as such, and multi-class sources give the class that was taken as positive.

Dataset	Domain	n	p	Minority	IR	Construction
turnover_russia	HR	1129	50	558	1.02	file: turnover.csv
saudi_attrition	HR	1191	100	515	1.31	file: Original Dataset of Employee Attrition.xlsx
job_change_ds	HR	5000	62	1247	3.01	file: aug_train.csv
ibm_hr	HR	1470	44	237	5.20	file: HR-Employee-Attrition.csv
hr_case_study	HR	4410	33	711	5.20	file: general_data.csv
healthcare_attr	HR	1676	40	199	7.42	file: watson_healthcare_modified.csv
bupa	benchmark	345	5	169	1.04	binary
sonar	benchmark	208	60	97	1.14	binary
credit_approval_australia	benchmark	690	15	307	1.25	binary
spambase	benchmark	4601	57	1813	1.54	binary
ionosphere	benchmark	351	33	126	1.79	binary
magic	benchmark	5000	10	1758	1.84	binary
tic_tac_toe	benchmark	958	9	332	1.89	binary
titanic	benchmark	2207	8	711	2.10	binary
credit_approval_germany	benchmark	1000	20	300	2.33	binary
phoneme	benchmark	5000	5	1467	2.41	binary
haberman	benchmark	306	3	81	2.78	binary
dna	benchmark	3186	180	765	3.16	one-vs-rest on class 2 of 3
adult	benchmark	5000	14	1196	3.18	binary
contraceptive_method	benchmark	1473	9	333	3.42	one-vs-rest on class 1 of 3
clean2	benchmark	5000	168	771	5.49	binary
segmentation	benchmark	2310	18	330	6.00	one-vs-rest on class 3 of 7
churn	benchmark	5000	20	707	6.07	binary
dermatology	benchmark	366	34	49	6.47	one-vs-rest on class 4 of 6
ecoli	benchmark	327	6	35	8.34	one-vs-rest on class 4 of 5
mfeat_karhunen	benchmark	2000	64	200	9.00	one-vs-rest on class 9 of 10
optdigits	benchmark	5000	62	493	9.14	one-vs-rest on class 8 of 10
satimage	benchmark	5000	36	486	9.29	one-vs-rest on class 4 of 6
pendigits	benchmark	5000	16	480	9.42	one-vs-rest on class 3 of 10
vowel	benchmark	990	13	90	10.00	one-vs-rest on class 7 of 11
analcatdata_authorship	benchmark	841	70	55	14.29	one-vs-rest on class 2 of 4
coil2000	benchmark	5000	85	298	15.78	binary
hypothyroid	benchmark	3163	25	151	19.95	binary
car_evaluation	benchmark	1728	6	65	25.58	one-vs-rest on class 3 of 4
letter	benchmark	5000	16	184	26.17	one-vs-rest on class 26 of 26
wine_quality_white	benchmark	4898	11	163	29.05	one-vs-rest on class 4 of 7
ann_thyroid	benchmark	5000	21	115	42.48	one-vs-rest on class 1 of 3
allhyper	benchmark	3771	27	79	46.73	one-vs-rest on class 2 of 4
yeast	benchmark	1479	8	30	48.30	one-vs-rest on class 7 of 9
dis	benchmark	3772	27	58	64.03	binary
allrep	benchmark	3772	27	34	109.94	one-vs-rest on class 1 of 4

3.2 The three conditions

Every comparison in this paper is between pipelines that differ in exactly one respect. The data, the preprocessing, the classifier, its hyperparameters, the resampler, its random seed, the split proportion and the split seed are all held fixed. The only thing that moves is where the resampler sits. In the leaked condition the test partition is drawn from the resampled pool, so it contains synthetic points and has a different class prevalence and size from the real test partition. This is deliberate: it is what the erroneous pipeline actually does, and reproducing its reported result faithfully is the point. In the correct and none conditions the test partition contains only real observations. The primary contrast is therefore an end-to-end workflow estimand. It combines training contamination, evaluation on synthetic observations and the prevalence shift induced by resampling; it should not be interpreted as the isolated causal effect of contaminated training data on a common test set. A complementary fixed-real-test experiment would be required to separate those channels. All features are standardised inside the pipeline, fitted on training data only in every condition, so that scaling is never the source of a difference.

3.3 A measure of the mechanism

If the account in Section 1 is right, the leaked condition should be recognisable not only by its outcome but by an intermediate quantity: how much of the test set the training partition has effectively already seen. We define the contamination rate as the fraction of held-out minority points whose nearest synthetic training point lies closer than half the median nearest-neighbour distance among real minority points. The reference distance is computed on the same standardised scale, so the measure is dimensionless and comparable across datasets. It is the mediating variable that turns an observation into an explanation, and it is what lets us say why the mitigation in Section 4.6 works.

3.4 The factorial

Classifiers are used at library defaults except where a default is plainly unsuitable for a study of this kind: the decision tree is capped at depth six and the forests at two hundred trees [45, 47, 49]. We do not tune, because tuning would add a second experimental layer whose folds and search procedure would themselves have to be made leakage-safe. The matched design therefore identifies the protocol difference conditional on these specified learners; it does not establish that the absolute scores, the variance ratio or the ranking of model families are invariant to hyperparameter optimisation. The seven families span linear, local, kernel, tree and boosted or bagged tabular learners, but exclude neural networks. A leakage-safe nested-tuning study and an MLP robustness condition are consequently tests of generality rather than results of the present factorial. Metrics recorded are minority-class F1, precision, recall, Matthews correlation, ROC-AUC and average precision; F1 is the headline because it is what the literature under study reports, and the others are used in Section 4.2 to check that the finding is not an artefact of one metric [17, 18, 46, 48].

Unless otherwise stated, bootstrap intervals in the descriptive summaries resample matched fits. They quantify stability within the assembled experimental record, not population-level uncertainty over a hypothetical random sample of datasets. The meta-regressions instead use standard errors clustered by dataset. Population-facing inference would require dataset-level or hierarchical resampling, with

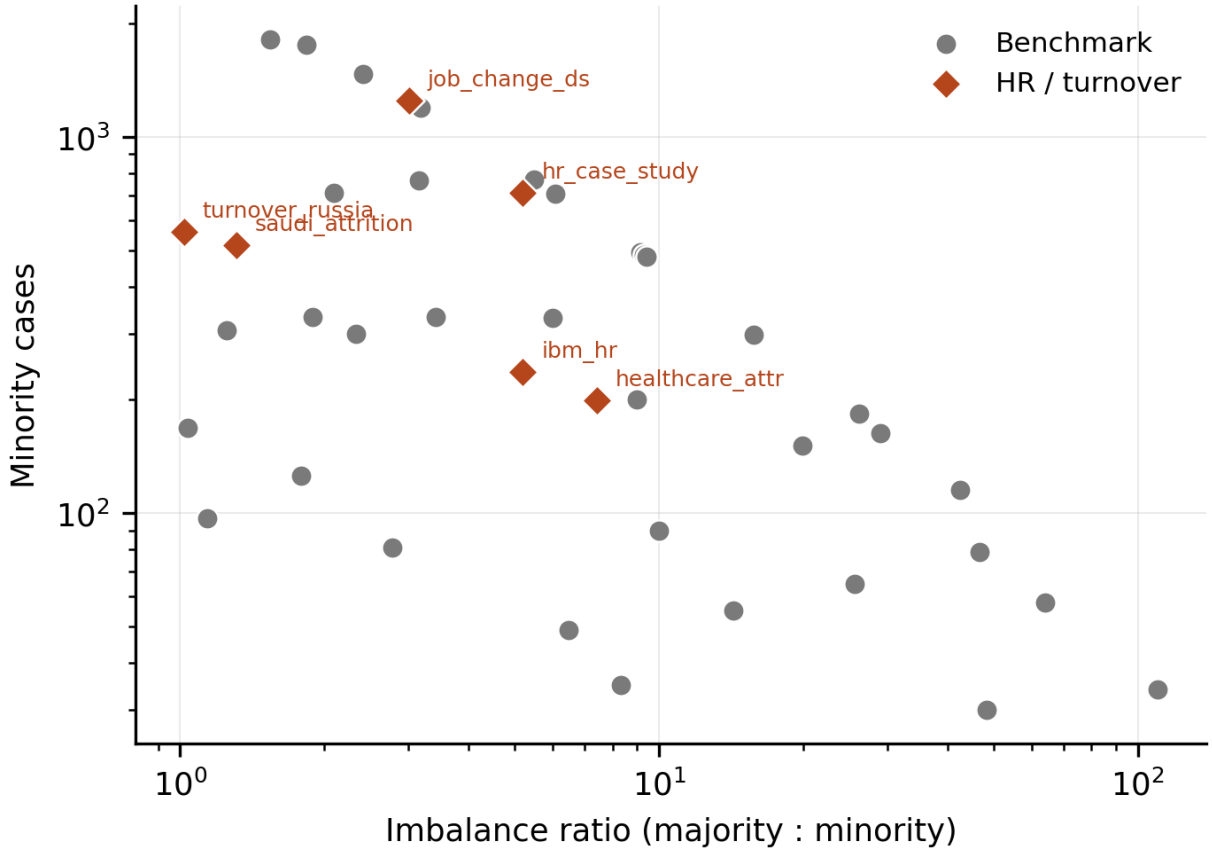


Figure 1: Coverage of the corpus. Each point is a dataset. The horizontal axis is the imbalance ratio and the vertical axis the number of minority cases, both on log scales. The employee-attrition datasets, labelled, occupy the mild-imbalance corner; the benchmark problems extend the range by two orders of magnitude, which is what allows the regularity in Section 4.2 to be stated as a function of difficulty rather than observed at a single point.

The three conditions. Everything except the position of the resampler is held identical, so any difference between conditions is attributable to that position alone.

Condition	Procedure	Role
leaked	Resample the entire dataset, then split 70:30	Reproduces the error as it appears in the literature
correct	Split 70:30, then resample the training partition only	The correct protocol
none	Split 70:30, no resampling	Baseline: is resampling helping at all?

The design. The seed dimension exists so that every comparison in Section 4 can be reported with a confidence interval.

Factor	Levels	Detail
Datasets	41	35 benchmark, 6 HR/turnover
Classifiers	7	LogReg, DecisionTree, 5-NN, SVM-RBF, RandomForest, GradientBoosting, XGBoost
Resamplers	4	SMOTE, Borderline-SMOTE, ADASYN, RandomOverSampler
Conditions	3	leaked / correct / none
Seeds	10	so that every comparison carries an interval, not a point
Model fits	25,620	25,340 completed; 11,200 matched leaked-correct pairs

the 41 datasets treated as the independent units.

3.5 Controlled simulation

Real datasets confound the things we most want to separate. Highly imbalanced problems tend also to be larger, higher-dimensional and more overlapping, because that is how such problems arise. A synthetic grid pulls the factors apart: we generate problems at five imbalance ratios, three dimensionalities, three class-separation levels and two sample sizes, and run the leaked and correct conditions on each with five classifiers and three seeds. The simulation is a supplement to the observational corpus and not a substitute for it; its role is to establish which of the co-varying factors is doing the work, and it is presented after the empirical results for that reason.

3.6 Meta-regression

To ask what governs the premium we regress each of the three quantities - the honest score, the leaked score and their difference - on the same set of descriptors: log imbalance ratio, log minority count, log feature count, log sample size, and indicator variables for classifier and resampler. Standard errors are clustered by dataset, since observations from one dataset are not independent. The comparison of interest is between the R^2 values: if the leaked score has stopped measuring the problem, the descriptors of the problem should explain much less of its variance than they explain of the honest score's.

3.7 The screening model and how it is validated

Each matched pair yields two labelled examples: a leaked result and an honest one. We train a gradient-boosted classifier to tell them apart using only features a reader can extract from a published paper—the reported F1, the imbalance ratio, the minority count, the feature count and the model family—and validate it leave-one-dataset-out, so that the estimate is never inflated by similarity between observations from the same dataset. A model that could only recognise the datasets it was trained on would be useless for the purpose, and leave-one-dataset-out is the validation that detects that failure. The construction contributes one leaked and one honest example per pair, imposing a 50:50 class prior. Its output is therefore a screening score, or a probability conditional on that experimental prior, not a calibrated posterior probability that an arbitrary published result is leaked.

3.8 Reproducibility

All datasets are public. The corpus builder downloads Group A and reads Group B from files with documented provenance, then writes a manifest that every downstream step reads. The experiment checkpoints after each dataset. All analysis flows into a single machine-readable summary, and every quantity in this manuscript is inserted from that summary at build time rather than typed, so the text cannot drift from the tables. Library versions at the time of the reported run were Python 3.11.15, NumPy 2.4.4, pandas 3.0.2, SciPy 1.17.1, scikit-learn 1.8.0 [19], imbalanced-learn 0.14.2 [20] and XGBoost 3.2.0 [21].

4 Results

4.1 The corpus behaves as intended

Table 1 and Figure 1 describe what was assembled. The imbalance ratio spans 1.02 to 109.9, minority counts 30 to 1813, and feature counts 3 to 180. The employee-attrition datasets sit at the mild end of the imbalance range, which is worth noting for what follows: the domain in which this error is most often committed is also the domain in which the honest score is highest, and therefore the domain in which the premium is smallest. The corpus is not a random sample of problems, and no claim here depends on treating it as one.

4.2 The invariance

This is the central result. Under the correct ordering, mean minority-class F1 is 0.673 with a standard deviation of 0.242. Under the leaked ordering, the mean rises to 0.896 but the standard deviation falls to 0.110. The honest score has 4.86 times the variance of the leaked one. Aggregating first to dataset means, which removes within-dataset noise, the ratio is 4.76: across the whole corpus, mean honest F1 ranges over 0.922 while mean leaked F1 ranges over only 0.389. Figure 2(a) shows the shape of it. Points lie above the diagonal, as expected. What is not expected is that they lie along a horizontal band rather than along a line parallel to the diagonal. A pipeline that merely added a bias would produce the latter. Figure 2(b) makes the same point against the variable that should matter most, and adds one we did not anticipate. As the imbalance ratio rises across the corpus, mean honest F1 falls from 0.744 in the least imbalanced quartile to 0.513 in the most imbalanced, which is what one expects of a measurement tracking a problem that is getting harder. Over the same range the leaked score does not merely stay flat: it rises, from 0.806 to 0.958. The correlation between log imbalance ratio and F1 is -0.268 under the correct ordering and 0.578 under the leaked one—opposite in sign, and larger in magnitude under leakage. Per decade of imbalance ratio the honest score falls by 0.122 while the leaked score rises by 0.120. The reason is the mechanism, and it belongs here rather than in Section 4.5. The more imbalanced a dataset is, the more synthetic points an over-sampler must manufacture to balance it, and so the more of the test set is present in the training partition in interpolated form. Difficulty and contamination rise together, and the second more than cancels the first. This is why the two scores diverge fastest exactly where the honest task is hardest, and it is what makes a high reported score on a highly imbalanced problem the specific thing to be suspicious of. Figure 2(c) states the consequence as a relation. The premium is $\text{premium} = 0.778 - 0.824 \times F1(\text{correct})$, with $R^2 = 0.795$. Aggregated to dataset means the fit is 0.794. Nearly all of the variation in how much a result is inflated is explained by how hard the problem was to begin with—which is another way of saying that the leaked score carries almost no information about the problem, so the difference between it and the honest score is almost entirely the honest score.

Table 2 checks the invariance against the design factors one at a time; within each factor we report the spread of condition means across levels. Across imbalance quartiles the honest score spreads 0.251 and the leaked score 0.152, in opposite directions. Across classifiers the two spread by comparable amounts, 0.147 and 0.108, but they rank the classifiers differently: the leaked score is highest for RandForest and lowest for LogReg, an ordering by capacity to memorise rather than by any property of the data. Across resamplers both scores spread by less than 0.013, so which over-sampler is

used is close to irrelevant to the artefact. What survives in the leaked score is variation from the estimator and from the amount of contamination the pipeline manufactured, and not from the question being asked.

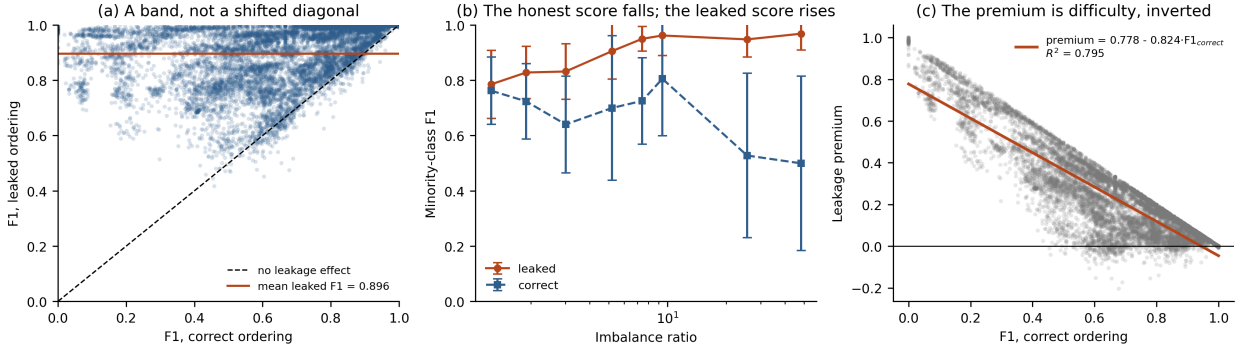


Figure 2: The invariance. (a) Leaked against honest F1 for all 11,200 matched pairs; the dashed line is equality and the horizontal line the mean leaked score. Points lie along a band, not along a shifted diagonal. (b) Both scores against the imbalance ratio, binned; error bars are one standard deviation. The honest score falls as the problem gets harder; the leaked score rises. (c) The premium against honest F1, with the fitted line: how much a result is inflated is almost entirely determined by how hard the problem actually was.

Table 2: Invariance across the design factors. Each row is the mean over all matched pairs at that level. The honest score varies strongly across imbalance levels and weakly across classifiers; the leaked score does the reverse. The 'no resampling' column is populated only for classifiers, where it is defined.

Factor	Level	Pairs	Leaked F1	Correct F1	No resampling	Premium
Imbalance ratio	1.02-2.3	2870	0.8062	0.7436		0.0626
Imbalance ratio	2.33-6.0	2800	0.8687	0.6699		0.1987
Imbalance ratio	6.00-14.7	2730	0.9560	0.7644		0.1916
Imbalance ratio	14.66-109.9	2800	0.9583	0.5135		0.4448
Classifier	5NN	1600	0.8826	0.6153	0.5981	0.2672
Classifier	DecTree	1600	0.8755	0.6377	0.6379	0.2378
Classifier	GradBoost	1600	0.9170	0.7192	0.6935	0.1978
Classifier	LogReg	1600	0.8279	0.5900	0.5451	0.2379
Classifier	RandForest	1600	0.9357	0.7237	0.6803	0.2119
Classifier	SVM-RBF	1600	0.9067	0.6862	0.5846	0.2205
Classifier	XGBoost	1600	0.9293	0.7370	0.7174	0.1923
Resampler	ADASYN	2590	0.9043	0.6633		0.2410
Resampler	Borderline	2870	0.8949	0.6736		0.2213
Resampler	RandomOS	2870	0.8923	0.6766		0.2156
Resampler	SMOTE	2870	0.8949	0.6766		0.2183
Domain	HR	1610	0.8260	0.5907		0.2353
Domain	benchmark	9590	0.9082	0.6865		0.2217

The finding is not an artefact of F1. Table 3 repeats the comparison for five other metrics. Matthews correlation shows a premium of 0.196 with a variance ratio of 1.6; average precision shows 0.229 and 6.91; ROC-AUC, which is insensitive to the class ratio and therefore the most conservative test available, still shows 0.056 and 1.76. The direction and the flattening survive the change of metric.

4.2b Where the F1 premium comes from: false positives versus false negatives

F1 conflates precision and recall, so a rise in F1 does not say which error fell. Reconstructing both from the stored precision and recall, on each protocol's own test split, false negatives per 100 actual minority cases are

$$100(1 - R),$$

and false positives per 100 actual minority cases are

$$100R\frac{1 - P}{P},$$

where P and R denote precision and recall, respectively.

These transformations are applied within each fitted model and then averaged across fits. Consequently, the reported mean error rates need not equal the same nonlinear transformations evaluated at the displayed mean precision and recall.

Across the corpus the leaked ordering cuts missed minority cases from 24 to 9 per 100 positives and cuts false alarms from 81 to 12 per 100 positives (Table 4). The mean F1 premium of 0.224 is the joint result of a precision lift of 0.248 and a recall lift of 0.149. Both errors fall; the larger movement is in the false-discovery rate.

That last fact needs a caveat, because it is easy to misread. The leaked test set is the rebalanced, partly synthetic partition created by resampling before splitting. A drop in FDR on that partition is not a drop in operational false positives on a live, imbalanced workforce. It is part of what the artefact is: the evaluation no longer asks the question a deployed model will be asked. The recall lift is closer to a genuine change in the fitted decision surface; the precision lift is largely a change in the test distribution. Table 4 therefore decomposes the reported F1. It does not claim that a leaked pipeline, once deployed, would raise precision by 0.25.

Table 3: The result is not specific to F1. 'Var ratio' is the variance of the honest score divided by the variance of the leaked score; values above one mean the leaked score is the flatter of the two.

Metric	**Leaked**	**sd**	**Correct**	**sd**	**Premium**	**Var ratio**
f1	0.8964	0.1099	0.6727	0.2424	0.2236	4.86
mcc	0.7934	0.2196	0.5977	0.2775	0.1957	1.60
pr_auc	0.9237	0.1036	0.6949	0.2724	0.2288	6.91
roc_auc	0.9332	0.0934	0.8771	0.1238	0.0561	1.76
precision	0.8900	0.1128	0.6424	0.2676	0.2476	5.63
recall	0.9063	0.1159	0.7568	0.2094	0.1495	3.26

Table 4: Precision, recall and reconstructed error rates. FN and FP are expressed per 100 actual minority cases on that protocol's test split. FDR = 1 − precision; FNR = 1 − recall. The leaked test set is class-balanced; the correct and none test sets retain the original prevalence.

Slice	**Protocol**	**Precision**	**Recall**	**F1**	**FDR**	**FNR**	**FN / 100 pos.**	**FP / 100 pos.**
All	correct	0.642	0.757	0.673	0.358	0.243	24.3	80.9
All	leaked	0.891	0.907	0.897	0.109	0.093	9.3	11.5
All	none	0.736	0.594	0.637	0.264	0.406	40.6	14.9
HR	correct	0.584	0.639	0.591	0.416	0.361	36.1	56.5
HR	leaked	0.821	0.835	0.826	0.179	0.165	16.5	18.7
Benchmark	correct	0.652	0.777	0.687	0.348	0.223	22.3	85.0
Benchmark	leaked	0.902	0.919	0.909	0.098	0.081	8.1	10.4

4.3 What the meta-regression shows, and what we expected instead

Table 5 reports the three regressions described in Section 3.6, and the result is not the one we set out to find. We expected the descriptors of a problem to explain the honest score well and the leaked score poorly. They do the opposite: dataset and algorithm descriptors explain 0.565 of the variance in the leaked score and only 0.175 of the variance in the honest one. Restricting the model to data descriptors alone—imbalance, minority count, dimensionality, sample size, with no algorithm terms—the gap is similar, 0.465 against 0.125. Algorithm terms alone explain little of either (0.102 and 0.050). We report this because it is what the data says, and because reading it correctly strengthens rather than weakens the argument. The resolution is in the sign. The leaked score is highly predictable from the imbalance ratio, but it moves the wrong way: its correlation with log imbalance is 0.578, positive, while the honest score's is -0.268. What the descriptors predict, when they predict the leaked score, is not how hard the problem is but how much contamination the pipeline manufactured - and imbalance is the main driver of that, because a more imbalanced problem needs more synthetic points to balance it. So the leaked score is not noise, and it is not a degraded copy of the honest score either. It is a well-behaved measurement of a different quantity. Predictability is therefore not evidence that a leaked number carries information about the problem; it carries information about the error. That is a sharper claim than the one we expected to make, and it is what makes the screen in Section 4.7 possible at all: a signature this systematic can be recognised. One further reading of Table 5 is worth stating. That the honest score is only weakly explained (0.175) by imbalance, sample size, dimensionality and model family is itself a useful result for anyone designing a benchmark study. Most of what makes a real classification problem hard is not captured by the descriptors a paper reports - class overlap and feature quality above all - which is precisely why the controlled simulation in the next section is necessary.

Table 5: Meta-regression of each outcome on dataset and algorithm descriptors, with standard errors clustered by dataset. R^2 is 0.175 for the honest score, 0.565 for the leaked score and 0.351 for the premium. The coefficients on the data terms are the ones to compare across columns.

Term	**Correct b**	**se**	**Leaked b**	**se**	**Premium b**	**se**
const	0.6856	0.2896	0.8863	0.1145	0.2007	0.2101
log_ir	0.4877	0.5862	0.8016	0.2529	0.3139	0.4422
log_nmin	0.7105	0.6977	0.7916	0.2819	0.0811	0.5372
log_p	0.1125	0.0736	0.0631	0.0219	-0.0494	0.0580
log_n	-0.7045	0.7131	-0.7977	0.2885	-0.0932	0.5484
clf_DecTree	0.0224	0.0239	-0.0070	0.0082	-0.0294	0.0194
clf_GradBoost	0.1039	0.0205	0.0344	0.0077	-0.0694	0.0179
clf_LogReg	-0.0253	0.0266	-0.0547	0.0163	-0.0294	0.0176
clf_RandForest	0.1084	0.0204	0.0531	0.0077	-0.0553	0.0195
clf_SVM-RBF	0.0709	0.0143	0.0241	0.0089	-0.0468	0.0082
clf_XGBoost	0.1216	0.0209	0.0467	0.0079	-0.0749	0.0193
sampler_Borderline	0.0067	0.0079	0.0003	0.0031	-0.0064	0.0070
sampler_RandomOS	0.0097	0.0077	-0.0023	0.0035	-0.0120	0.0072
sampler_SMOTE	0.0096	0.0078	0.0003	0.0031	-0.0093	0.0069

4.4 The simulation isolates the cause

The corpus cannot separate imbalance from the things that travel with it. The simulation can. Across 1350 synthetic fits, the same asymmetry appears: honest F1 has 8.38 times the variance of leaked F1. Varying the imbalance ratio alone, with dimensionality, class separation and sample size held fixed, honest F1 falls from 0.773 to 0.249 - a drop of 0.525 - while leaked F1 moves by 0.042. Varying dimensionality alone moves the honest score by 0.153 and the leaked score by 0.026. Varying class separation moves them by 0.296 and 0.087. Figure 3 shows all three comparisons side by side. This settles the direction of the causal story. The premium grows with imbalance not because leakage helps more when data is scarce, but because the honest score has further to fall. The leaked score barely moves at all. That is a different claim from "leakage inflates results", and it is the one the data supports.

4.5 The mechanism, measured

The contamination rate defined in Section 3.3 averages 0.637 across the corpus and ranges from 0.000 to 1. Its correlation with the log imbalance ratio is 0.724, which identifies it as the channel through which imbalance raises the leaked score rather than lowering it. It correlates with the premium at r = 0.522 and with the leaked score at r = 0.442. Contamination is a genuine mediator but not a complete one, and it is worth being precise about that. Entering it alongside honest difficulty in a regression on the premium moves the coefficient on difficulty only from -0.830 to -0.753, about 9.3% of the total. That is what one should expect from the definition: the contamination rate measures how much of the test set has been copied into training, and not what a given model does with the copies. How much a copy is worth depends on the estimator's capacity to memorise, which is why the classifier remains the second source of variation in Table 2. The resampler comparison is the sharpest test the design offers, and it separates the two halves of the mechanism cleanly. RandomOverSampler duplicates existing minority points rather than interpolating new ones, so under the leaked ordering it puts exact copies of test points into training rather than near copies. It produces by far the highest contamination of any resampler in the corpus, 0.785 against 0.464 for Borderline-SMOTE, which is exactly what the definition predicts. Its premium, however, is not the largest: it is 0.216, against 0.241 for ADASYN. More contamination does not mean a proportionally larger artefact, because a duplicated point teaches a model less than an interpolated neighbourhood does. The four resamplers differ from one another by less than 0.012 in leaked F1 - which is the substantive point. What matters is that synthetic points cross the split at all, not how they were made.

4.6 Partial resampling as a safeguard

If the artefact comes from synthetic points crossing the split, resampling to less than full balance should shrink it, because there are fewer such points to cross. Table 6 and Figure 4(b) confirm the direction across the whole corpus, and quantify a reduction that is real but partial. At full balance the premium is 0.221. At a target minority prevalence of 0.25 it is 0.166, a reduction of 25%, and at 0.15 it is 0.155. The contamination rate falls in step, from 0.696 to 0.517, which is what the mechanism predicts and what makes the reduction interpretable rather than merely convenient. A quarter is not most of it. Partial resampling attenuates the artefact; it does not remove it, and a study that resamples to 0.15 before splitting still reports a number inflated by 0.15. Anyone

hoping for a setting that makes the ordering safe should read Table 6 as ruling that out. We are not recommending partial resampling as a way to make the error safe. The error should be fixed, not attenuated. The finding matters for a different reason: it gives a diagnostic signature. A pipeline whose reported performance changes sharply with the resampling ratio, in a way that cannot be explained by the classifier, is a pipeline worth inspecting. It is also a reason to doubt full rebalancing on its own merits, which theoretical analyses, reviews and broad empirical comparisons have argued from other directions [12, 13, 14, 35, 38, 39, 40].

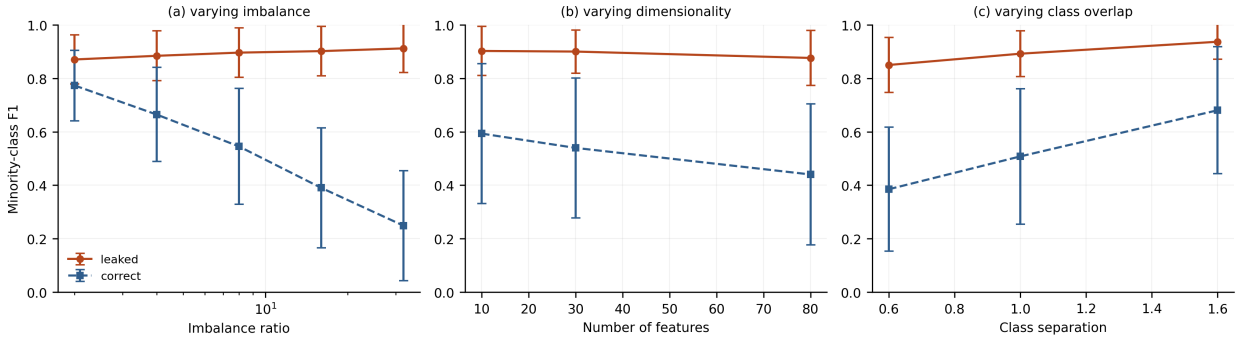


Figure 3: Controlled simulation. Each panel varies one factor with the others held fixed; error bars are one standard deviation across cells and seeds. The leaked curve is close to flat in all three panels while the honest curve responds to each factor, which is what identifies problem difficulty rather than leakage magnitude as the thing that moves.

Table 6: Premium as a function of how far SMOTE is asked to rebalance the data, across the whole corpus. Confidence intervals are bootstrap percentile intervals over matched fits.

Target prevalence	Fits	Premium	sd	Leaked	Correct	Contamination	CI low	CI high
0.15	162	0.1548	0.1799	0.7579	0.6032	0.5167	0.1283	0.1829
0.25	216	0.1656	0.1849	0.8138	0.6482	0.6007	0.1427	0.1896
0.4	261	0.1883	0.1990	0.8184	0.6301	0.6460	0.1646	0.2137
0.6	315	0.2006	0.2051	0.8434	0.6427	0.6721	0.1783	0.2231
0.8	333	0.2211	0.2103	0.8743	0.6533	0.7080	0.1981	0.2425
1.0	369	0.2209	0.2180	0.8766	0.6558	0.6959	0.1984	0.2441

4.7 Screening a reported result

The final result asks whether the signature is stable enough to be recognised from outside. Using only the reported F1, imbalance ratio, minority count, feature count and model family, and validating leave-one-dataset-out over 22,400 labelled results, the screen separates leaked from honest results with AUC 0.795. Accuracy at the natural threshold is 0.697 and the Brier score is 0.197. The gap between two ways of scoring the same screen is instructive. Within a dataset, where the task is to say which of two results on the same problem was the leaked one, per-dataset AUC has median 0.987 and exceeds 0.9 on 27 of 41 datasets; it falls below chance on 1. Pooled across datasets, where the task is to judge one reported result in isolation without knowing what an honest result on that problem would have looked like, the AUC is the 0.795 quoted above. The screen is therefore very good at the easy version of the question and moderately good at the hard one - and the hard one is what a reviewer actually faces. We report the pooled figure as the headline for that reason. An ablation makes the interpretation concrete. Removing the reported F1 and leaving only the dataset descriptors drops the AUC to 0.500, close to chance, as it must be - the descriptors alone carry no information about which condition produced a result. The signal is entirely in the relationship between the reported score and the difficulty of the problem it was reported on, which is the relationship Section 4.2 showed leakage destroys.

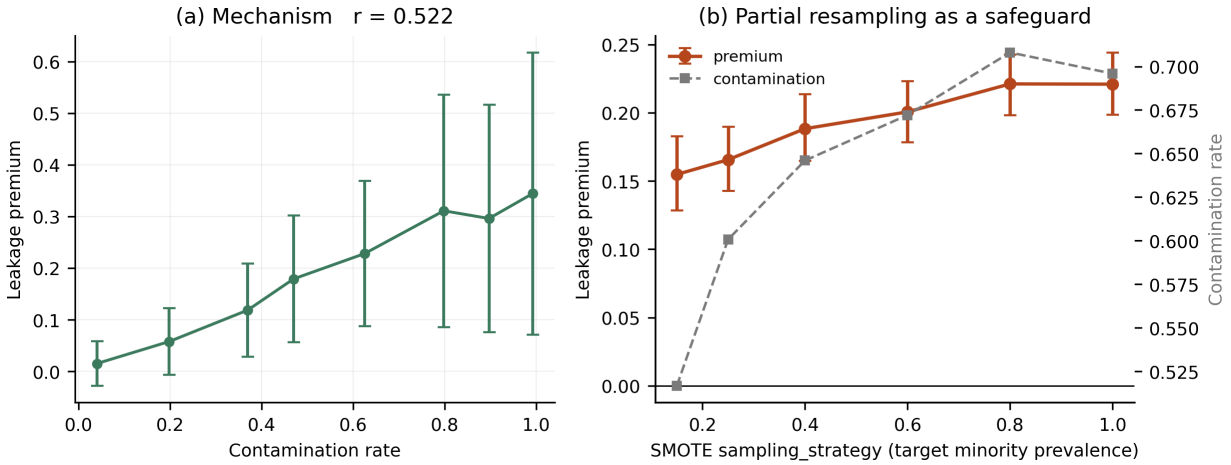


Figure 4: (a) The premium against the contamination rate, binned into deciles; the mechanism is measurable, not merely plausible. (b) The premium and the contamination rate as functions of the target minority prevalence. Both fall together, which is the signature of a single cause.

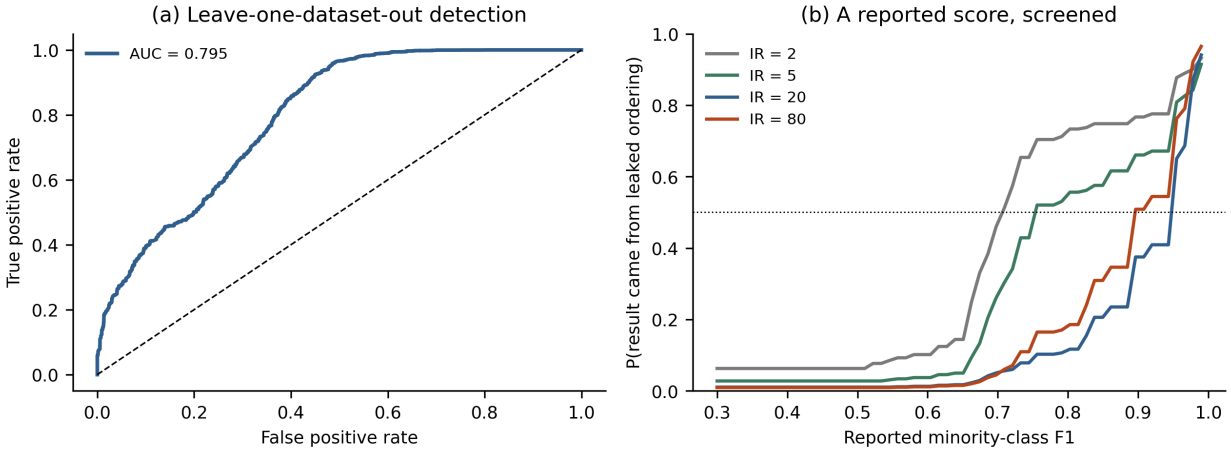


Figure 5: (a) Leave-one-dataset-out ROC for the screen. (b) The screen as a reader would use it: probability that a reported result came from the leaked ordering, as a function of the reported F1, at four imbalance ratios, holding the other inputs at corpus medians. The curves shift left as the problem gets harder: on a hard problem, a moderately high score is already unusual.

4.7b What the screen attends to

An ablation already showed that removing the reported F1 collapses leave-one-dataset-out AUC to chance. SHAP values [55] on a 3,000-row sample of the paired corpus make the same point at the level of individual inputs (Figure 6). Mean absolute SHAP is dominated by the reported F1 (1.88), followed by $\log_{10}$ imbalance ratio (0.57). Dimensionality, minority count and model-family dummies are smaller; logistic regression is the only family with a noticeable residual, consistent with a linear model being less able to memorise interpolated near-duplicates.

The screen is therefore not a classifier of datasets and not a classifier of algorithms. It is a classifier of pairs $(F1, \text{difficulty})$: a high score on a hard problem is the leaked signature, and everything else is second-order. That is why a reviewer can use it from quantities a paper already reports.

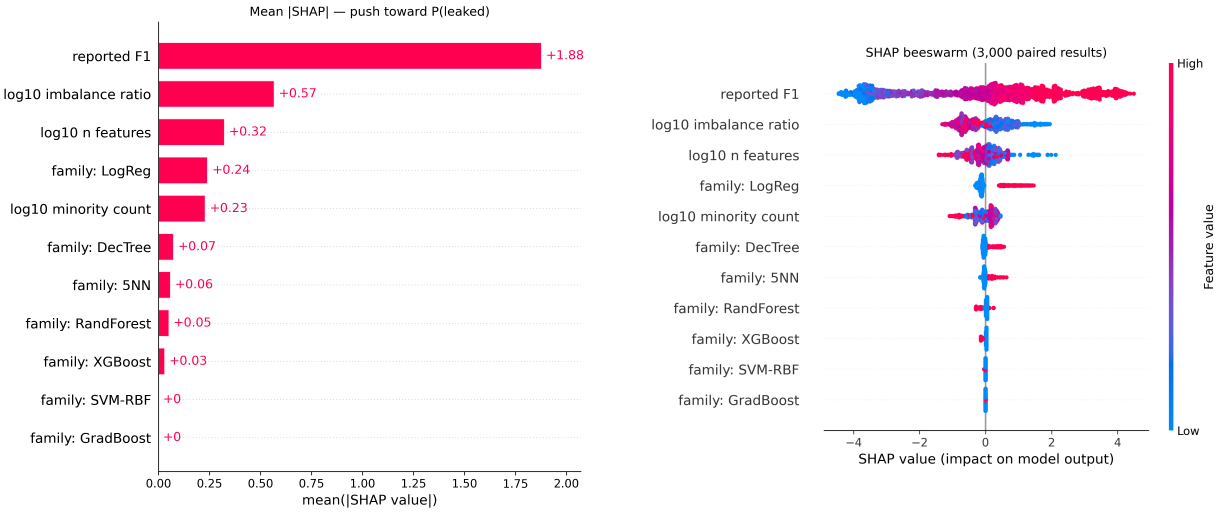


Figure 6: SHAP explanation of the screening model. (a) Mean |SHAP| over 3,000 paired leaked/honest results. (b) Beeswarm: each point is one result; colour is the feature value. Positive SHAP pushes toward $P(\text{leaked})$.

We stress what the screen is and is not. It returns a score for how characteristic a number of that size, on a problem of that difficulty, is of the leaked ordering under the balanced training construction. It is not a prevalence-calibrated probability and is not evidence of error in any individual paper, because it sees only what the paper reported. Its use is to tell a reviewer or an editor when it is worth asking the authors a specific question: where does the resampler sit relative to the split?

4.8 What the published literature reports

Table 7 places our measured bands against results reported in the recent employee-attrition literature, a domain where over-sampling is near-universal and where the IBM HR benchmark gives an unusually clean comparison because so many studies use the same data. Recent predictive, comparative and bibliometric studies confirm the continuing scale of this literature and its reliance on imbalance-handling methods [28, 41, 42, 43, 52, 53, 54]. We classify each study only by what it says about its own protocol—never by what our screen guesses—into three groups: resampling confined to

training data, resampling applied before splitting, and no statement either way. The pattern is stark. The 2 studies that state resampling was confined to the training data report minority-class F1 between 0.439 and 0.504 [22, 23]. The study that states resampling preceded the split reports 0.9158 [24]. The 4 results that do not state the ordering report between 0.659 and 0.990 [25, 26, 27]. Our own measurements on the same dataset, where the ordering is known because we controlled it, put honest F1 at 0.434 and leaked F1 at 0.877. We draw one conclusion from this and decline to draw another. The conclusion we draw is that the ordering is frequently not reported, and that where it is reported the difference in outcomes is large enough that a reader cannot treat the omission as immaterial. The conclusion we decline to draw is that any particular unreported study is in error. Several other explanations are available - different preprocessing, different splits, different metric definitions, a macro-averaged rather than minority-class F1 - and none of them can be ruled out from the outside. That is the point: from the outside, nothing can be ruled out, which is why the ordering belongs in the methods section.

Table 7: Reported minority-class F1 in recent employee-attrition studies, grouped by what each paper states about where resampling sits relative to the split. Values are as reported by the authors for the model they nominate as best. 'Not stated' means the paper does not describe the ordering, not that it is incorrect.

Study	**Venue**	**Dataset**	**Best model**	**Reported F1**	**Stated protocol**
Hamja et al. (2025)	Annals of Data Science	IBM HR	GradientBoosting	0.9900	not stated
Nassreddine et al. (2026)	Computers	IBM HR	Ensemble (ROS)	0.9774	not stated
Baydili & Tasci (2025)	Systems	IBM HR	GAN-Transformer	0.9158	before-split
AL-Ali et al. (2026)	Scientific Reports	IBM HR	AdaBoost	0.7969	not stated
AL-Ali et al. (2026)	Scientific Reports	Job change	HistGradientBoosting	0.6594	not stated
Alyousef et al. (2026)	Information	IBM HR	LogisticRegression	0.5042	train-only
Cavescu & Popescu (2026)	Information	IBM HR	LogisticRegression	0.4390	train-only

5 Discussion

5.1 Why a literature converges on numbers that look alike

Sections of the applied literature on imbalanced classification report headline scores in a narrow high band across very different datasets, feature sets and model families [28, 33, 34, 37, 38, 41, 42, 43]. Read as measurements, that convergence is puzzling: problems that differ in difficulty should produce results that differ in quality. Section 4.2 supplies an explanation that requires no assumption of bad faith. Under the leaked ordering the reported number is not a measurement of the problem, and quantities that do not measure the problem do not vary with it. A band is exactly what one should expect. The explanation is testable and we have tested part of it: the position of the band in our corpus depends on the classifier, and its width is a small fraction of the width of the honest distribution. It also makes a prediction that a reader can check against any literature they know well. Where a body of work reports scores that vary little across datasets of visibly different difficulty, the ordering of resampling and splitting is the first thing to look at.

5.2 Leakage as substitution, not inflation

The most useful thing this study changes is a description. 'Over-optimistic' suggests a bias term: a true value plus an increment, so that a leaked result still carries information about the truth and could in principle be corrected. More broadly, such underspecification of an evaluation procedure is a recognised threat to the credibility of machine-learning results [51]. Our results say otherwise. The leaked score correlates with the honest score at only 0.389, moves by 0.389 across a corpus over which the honest score moves by 0.922, and what movement it has runs in the wrong direction. There is little to correct, because what is there is a measurement of something else: of how much contamination the pipeline produced. This also explains why the premium is so predictable. If the leaked score were the honest score plus noise, the difference would be noise. Because the leaked score is close to constant, the difference is essentially the honest score with its sign flipped - which is what the fit in Figure 2(c) shows, and why R^2 is 0.795 rather than something modest.

5.3 What follows for review and editing

Reviewers are not usually able to run the code, and often cannot see it. What they can do is ask a question whose answer is one sentence long, and notice when the answer is missing. Three concrete practices follow from our results.

- Require the ordering to be stated. A methods section should say where resampling occurs relative to the split, and where it occurs relative to cross-validation folds. This is one sentence and it is currently absent from a large share of applied papers.
- Treat a high score on a hard problem as a question, not a finding. Figure 5(b) makes the trade-off explicit: on a problem with an imbalance ratio near 109.9, a reported F1 that would be unremarkable at low imbalance is already unusual.
- Ask for the no-resampling baseline. It costs the authors one extra run, and in 34.7% of our matched comparisons it was the stronger of the two correctly-evaluated options. A paper that reports it has answered the question of whether resampling helped at all.

5.4 What follows for practice

For practitioners the operative finding is Section 4.6. Full rebalancing is not required by anything, and it maximises the number of synthetic points in play. Contemporary reviews and method proposals increasingly emphasise data geometry, local quality and overlap rather than full balance as an automatic target [35, 36, 37, 39, 40]. Targeting a minority prevalence of 0.25 rather than 1.0 cut the premium by 25% in our corpus and reduced contamination in step. That is not a licence to resample before splitting; it is an argument that the default of full balance is a choice that costs something and should be justified.

5.4b An accounting illustration: what a leaked F1 would license a retention budget to do

The decomposition in Section 4.2b has a direct cost reading, which we give as an illustration on IBM HR ($n = 1{,}470$, 237 leavers), not as a forecast for any employer. Suppose a one-cycle programme flags predicted leavers for a retention action costing C_r per person, and that replacing a leaver costs $C_t = \$15{,}000$. True positives, false positives and false negatives then follow from the protocol's reported precision and recall. Net savings against a do-nothing baseline, and the share of spend wasted on people who were not going to leave, are plotted against C_r in Figure 7.

At $C_r = \$3{,}000$ the leaked evaluation implies a false-positive bill about \$262,000 smaller than the honest evaluation of the same models on the same data. That is the amount a team would under-provision for false positives if it treated a leaked F1 of 0.88 as an operational rate. The leaked curve is a budget, not a realised outcome: once the model is scored on new employees who were never in the resampling pool, the honest rates are the relevant ones. We do not vary C_t or intervention success; changing either rescales the figure and does not reverse the ranking of the two protocols on the false-positive bill.



Figure 7: IBM HR, illustrative costing. (a) Net savings versus do-nothing as a function of retention-action cost C_r. (b) Spend wasted on false positives. Solid: correct ordering; dashed: leaked; dotted: no resampling. Replacement cost is held at \$15,000. Precision and recall under the leaked protocol are measured on a rebalanced test set. The figure asks what those reported rates would license a budget to do; it does not simulate deployment on a live, imbalanced workforce.

5.5 Limitations

- Our corpus is assembled, not sampled. It spans a wide range of imbalance ratios by construction, and the distribution of problems within that range is not the distribution of problems in any field. Statements about the shape of the relationship are supported; statements about the typical magnitude in a given discipline are not.
- Multi-class benchmarks are binarised one-versus-rest. That is the standard construction for imbalanced benchmark suites and it is how the sources we drew on were built, but it produces problems whose minority class is a coherent category, which may be easier than a genuinely heterogeneous minority.
- Classifiers are untuned and neural networks are not represented. Matching the learner across protocols protects the within-configuration contrast, but hyperparameter optimisation can alter capacity, honest-score dispersion and therefore the reported variance ratio. Leakage-safe nested tuning and an MLP condition are needed before generalising that ratio beyond the specified model configurations.
- Datasets are capped at 5,000 rows. This prevents the largest datasets from dominating the factorial but restricts the target population of the measurement. Because oversampling leakage can vary with sample size, a within-dataset sensitivity analysis at multiple caps is needed before extrapolating to enterprise-scale data.
- The primary leaked and correct conditions do not share an identical real test set. The measured premium faithfully reproduces the erroneous end-to-end workflow, but combines training contamination with synthetic test observations and a shifted test prevalence. A fixed-real-test experiment is required to isolate the training-contamination component.
- Descriptive bootstrap intervals over matched fits do not make the 11,200 comparisons independent. Dataset-level bootstrap or hierarchical modelling would be required for population-level confidence intervals; clustered standard errors are used only in the meta-regressions reported here.
- The contamination threshold—one half of the median real-minority nearest-neighbour distance—is interpretable but chosen rather than estimated. Its absolute level should be tested against alternative thresholds before being treated as a portable quantity.
- The screen is trained on our corpus and inherits its coverage. Applied to a problem far outside the fitted ranges it extrapolates, and the released implementation says so in its output rather than returning a bare number.
- The screen is trained on an artificially balanced leaked-versus-honest sample. Its AUC measures ranking under leave-one-dataset-out validation, but its numeric outputs require recalibration before they can be interpreted under the unknown prevalence of leakage in published work.
- We measure one leakage mechanism. Preprocessing fitted on the full dataset, feature selection before splitting, and duplicate records across the split are separate errors with separate signatures, and nothing here estimates their magnitude.

5.6 A reporting checklist

The following six items would, if reported, let a reader evaluate any study that applies resampling to imbalanced data. None requires extra experiments.

1. Where the resampler sits relative to the split, and relative to cross-validation folds.
2. The class balance of the test set as evaluated: real observations only, or resampled.
3. The number of minority cases before resampling, not only the ratio.

4. The score without resampling, as a baseline.
5. The resampling target prevalence, if it is not full balance.
6. Precision and recall (or TP/FP/FN counts), not F1 alone, so a reader can see whether a high score reflects fewer misses, fewer false alarms or a change in test-set prevalence.

6 Conclusion

Within the evaluated capped tabular regime, resampling before splitting inflates minority-class F1 by 0.224 on average across 41 datasets, seven classifier families and four over-samplers. That number is the least interesting result in this paper. The consequential one is that the inflation is not additive: under the leaked ordering the reported score stops varying with the difficulty of the problem, retaining 4.86 times less variance than the honest score, and what variation remains runs against difficulty rather than with it. The premium is therefore predictable from difficulty alone, with R^2 0.795, and the artefact leaves a signature stable enough to be ranked in results one did not produce, at AUC 0.795 under leave-one-dataset-out validation. These conclusions concern the end-to-end erroneous workflow; separating contaminated training from the accompanying synthetic, prevalence-shifted test set requires a fixed-real-test robustness experiment. Two things follow that a practitioner can act on. Resampling to partial rather than full balance cuts the artefact by 25% and gives a diagnostic handle on it, though it does not remove it. And a reported result can be screened from what papers already publish, which turns a general warning about leakage into a specific question about a specific study. We release the screen, the corpus builder and the full experimental record so that both can be checked and extended.

Data and code availability

All datasets used are public. The corpus builder, the experiment, the analysis and the manuscript build script are released together, along with the complete record of every model fit reported here and the machine-readable summary from which every number in this text was inserted. The code and supporting materials are available from the GitHub repository `https://github.com/occbuu/leakage-premium` and are archived on Zenodo under DOI 10.5281/zenodo.22196384.

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